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A New Class of Superhalogen Based Anion Receptor in Li-ion Battery Electrolytes

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ABSTRACT: In the search for new additives in Li-ion battery electrolytes especially for LiPF₆ and LiClO₄, we have theoretically designed boron based complexes by coupling with different heterocyclic ligands. The validation of the formation of modeled compounds involves reproduction of available experimentally reported absolute magnetic shielding and chemical shift values for different boron-complexes. As compared to the commonly used tris (pentafluorophenyl) borane, our designed

1 compounds suggest that the complexes like $B[C_2HBNO(CN)_2]_3$, $B[C_2HBNS(CN)_2]_3$ and
2 $B[C_4H_3BN(CN)_2]_3$ are promising additives.
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23 INTRODUCTION

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26 In this modern era, if we think about portable electronic devices, the most appropriate
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29 alternative is perhaps the Li-ion battery¹⁻⁴. It basically consists of a cathode, anode and an
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32 electrolyte, mostly anhydrous in nature. The Li-ion battery is proved to be a better candidate⁵⁻⁶
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35 for an instant power source. Nevertheless, extensive ongoing researches are well under way
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38 in order to improve the performance in terms of safety and recyclability towards achieving
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41 higher energy and power density. The performance of Li-ion battery can be enhanced through
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44 modifications in the cathode, anode and the electrolyte. The electrolyte usually consists of a Li
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47 salt dissolved in a solvent like a linear or cyclic carbonate⁷⁻¹⁰. Upon dissociation, electrolytes
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50 not only enhance the Li-ion conduction but also form a solid electrolyte interface (SEI) at the
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57 electrodes. This eventually hampers¹¹⁻¹³ the electrolyte decomposition during the progress of
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the electrochemical reaction. At this point, the addition of functional additives¹⁴ to the electrolytes improve the thermal stability and cyclability of a Li-ion battery. The overall performance of battery is improved in two ways. Firstly, the additives enhance the Li-ion movement and secondly, they control the components of the SEI to prevent hindering the ion mobility. A common and popular additive used in such electrochemical reactions is tris(pentafluorophenyl) borane (TPFPB)¹⁵. TPFPB actually acts as an anion receptor which not only coordinates with the anionic counterpart of the electrolyte but also dissolves the resistant part of the SEI¹⁶⁻¹⁸. Owing to their high ionic conductivity, LiPF_6 and LiClO_4 based electrolytes are commonly used in Li-ion batteries. However, both the salts have disadvantages. LiPF_6 has low thermal stability and it undergoes dissociation to produce Li^+ and PF_6^- ions. The PF_6^- ion further dissociates into F^- ion and PF_5 . The Li^+ ion coordinates with the F^- ion to produce LiF which gets precipitated. Therefore, the number of free Li^+ ions in solution decreases and hence the conductivity of the electrolyte falls down. Additionally, PF_6^- initiates the polymerization of cyclic carbonate (the solvent) and hence degradation of the electrolyte occurs. On the other hand, LiClO_4 dissociates slowly to generate Li^+ ion but it has a safety issue arising out of the oxidation environment during charging. TPFPB acts as an excellent candidate to overcome the problems arising from LiPF_6 and LiClO_4 . It has been reported that for LiPF_6 , TPFPB coordinates with PF_6^- ion thus enhances the Li^+ ion dissolution followed by

an augmentation of the Li^+ transference number. Addition of TPFPB also increases the dissociation of LiClO_4 . The interaction between TPFPB and the anions like PF_6^- and ClO_4^- can be explained by a Lewis acid-base approach. The electron-withdrawing C_6F_5 ligands coordinated to the boron centre in TPFPB makes the B center electron deficient. Thus, the electron-deficient B centre will interact with an anion to form a Lewis acid-base type TPFPB-anion complex that will help to restrict the migration of the anion. A recent article of Giri and coworkers¹⁹ has discussed about the modeling of boron-based organic heterocyclic superhalogens as better anionic counterparts which upon coupling with Li can act as electrolytes in Li-ion batteries. Inspired by this rationale, we have made an attempt to design superhalogen-based anionic receptors containing an electron-deficient boron site which can be used as an alternative to TPFPB in Li-ion batteries. In this context, it may be noted that the Superhalogens (SH) are a category of molecules having electron affinities more than those of group 17 elements (halogens), mainly Chlorine (Cl) which has the highest electron affinity in the periodic table. Research in the past 10 years has greatly expanded the scope of superhalogens by suitably choosing the core as well as the ligand atoms.²⁰⁻²³

COMPUTATIONAL DETAILS

Geometry optimization has been carried out by employing wB97XD²⁴ level of theory and 6-311+G(d,p) as the basis set. Vibrational frequency analysis has been performed in the same level and basis set to ensure the ground state geometries. The gas phase optimized geometries were further taken to perform Single point calculation in solvent phase. We have used ethylene carbonate as a solvent in PCM²⁵ (polarizable continuum model) model. The dynamical stability of the proposed molecules has been investigated by performing an Atom Centered Density Matrix Propagation (ADMP)²⁶ molecular dynamics studies. We have generated a 1000 fs trajectory at room temperature with time step of 1 fs. For each case, the gas phase optimized geometry has been considered as the initial geometry. We have also performed ¹¹B NMR studies with reference to BF₃-OEt₂ theoretically by using wB97XD, B3LYP level of theory and 6-311+G(d,p), 6-311+G(2d,p) as basis sets to validate our results with the experimental findings. All the calculations are performed by Gaussian G09²⁷ software.

RESULT AND DISCUSSIONS

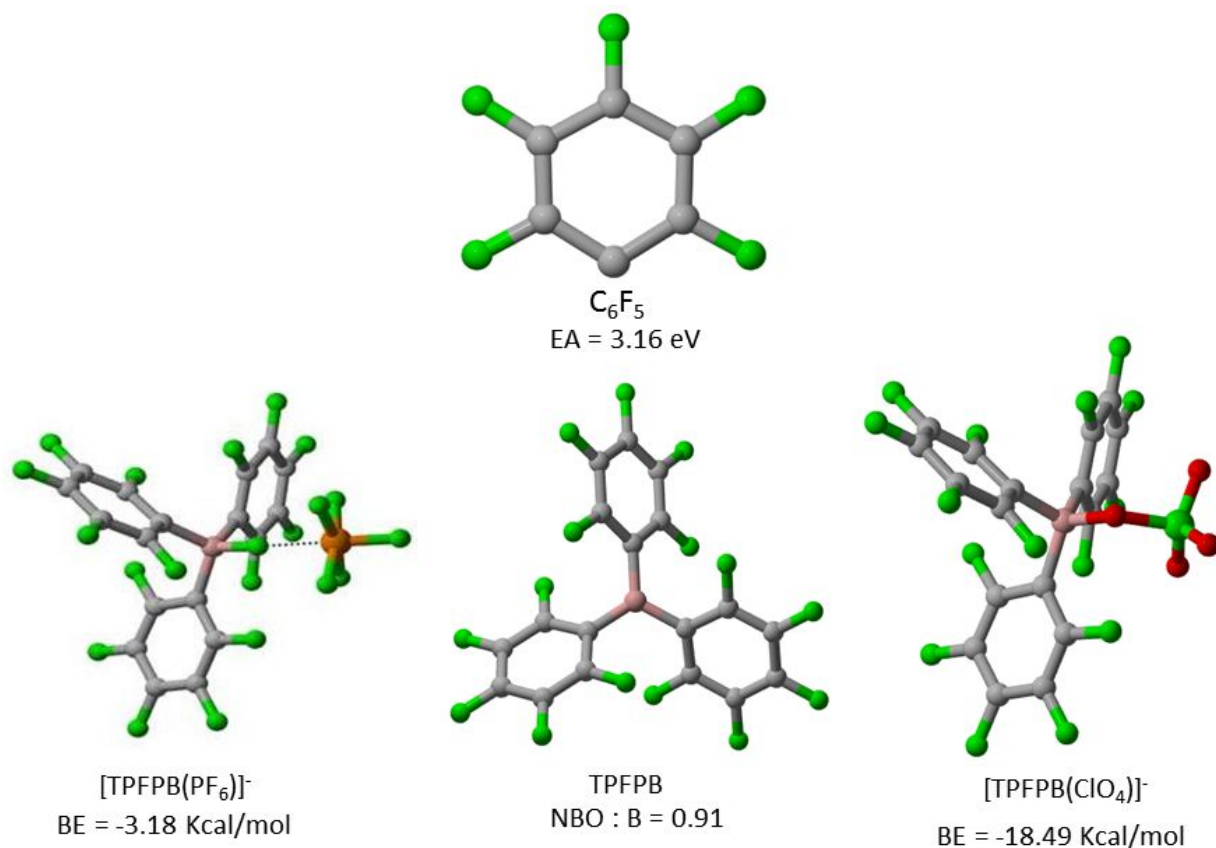


Figure 1. Optimized geometries of C_6H_5 , TFPB, ClO_4^- and PF_6^- trapped TFPB with binding energy (BE) and NPA charge of center B atom.

Figure 1 depicts the optimized molecular geometry of C_6H_5 ligand, TFPB and the associated Lewis acid-base complexes with the ClO_4^- and PF_6^- anions. The C_6F_5 ligand with an electron affinity (EA) value of 3.16 eV renders the electron-deficient B-site in TFPB with a formal positive NPA (Natural Population Analysis) charge (0.91 |e|) as obtained from NBO (Natural Bond Orbital) analysis. The resultant complexes with the ClO_4^- and PF_6^- anions are stable with negative binding energy values of -18.49 and -3.18 kcal/mol respectively. The

optimized molecular geometries of some representative superhalogens are given in Figure 2 and their associated electron affinity (EA) values, NPA charges are given in Table 1.

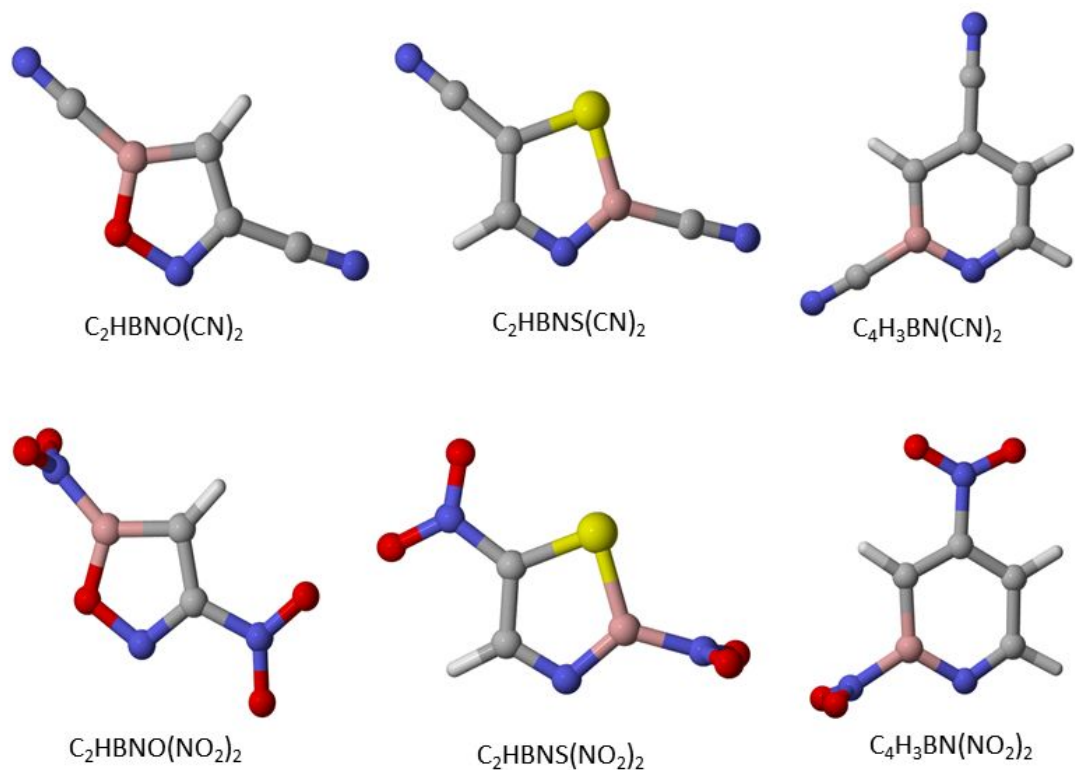


Figure 2. Optimized geometries of CN and NO₂ substituted individual superhalogen molecules.

Table 1. Electron affinities along with NPA charges on boron atom of individual superhalogen molecules.

S. No.	Molecules	Electron affinity (eV)	NPA charge on Boron
1	$C_2HBNO(CN)_2$	4.01	0.759
2	$C_2HBNS(CN)_2$	4.04	0.564
3	$C_4H_3BN(CN)_2$	4.19	0.686
4	$C_2HBNO(NO_2)_2$	4.32	0.645
5	$C_2HBNS(NO_2)_2$	4.48	0.764
6	$C_4H_3BN(NO_2)_2$	4.37	0.891

The calculated electron affinity values (ranges from 4.01 to 4.48 eV) of these molecules suggest that all the designed complexes are superhalogenic in nature. We have further found that CN substituted complexes are less superhalogenic than their NO₂ substituted counterpart. This probably due to more electron withdrawing nature of NO₂ than CN. We have also found the existence of positive boron centers in all the complexes as comparable to TFPB. All such results suggest that our designed complexes can act as Lewis acids and are capable to trap anions to form stable complexes. The optimized structures of the anion-trapped (ClO₄⁻ and PF₆⁻) complexes with CN-substituted and NO₂-substituted superhalogens are shown in Figure 3 and 4, respectively.

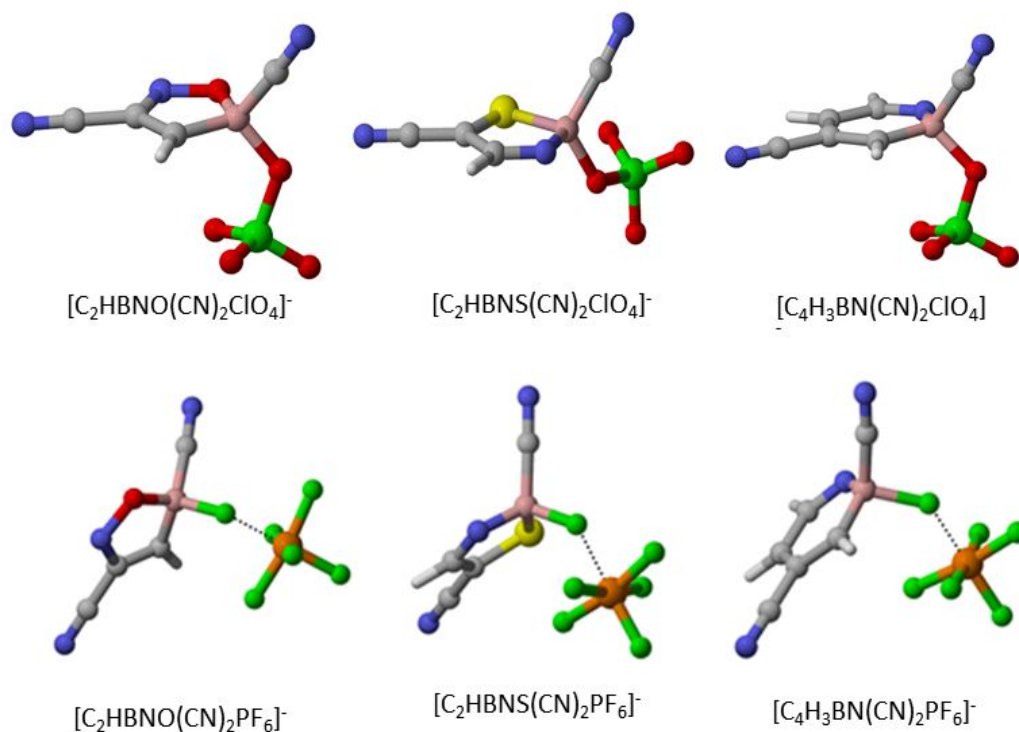


Figure 3. Optimized geometries of ClO₄⁻ and PF₆⁻ trapped by CN substituted Superhalogens.

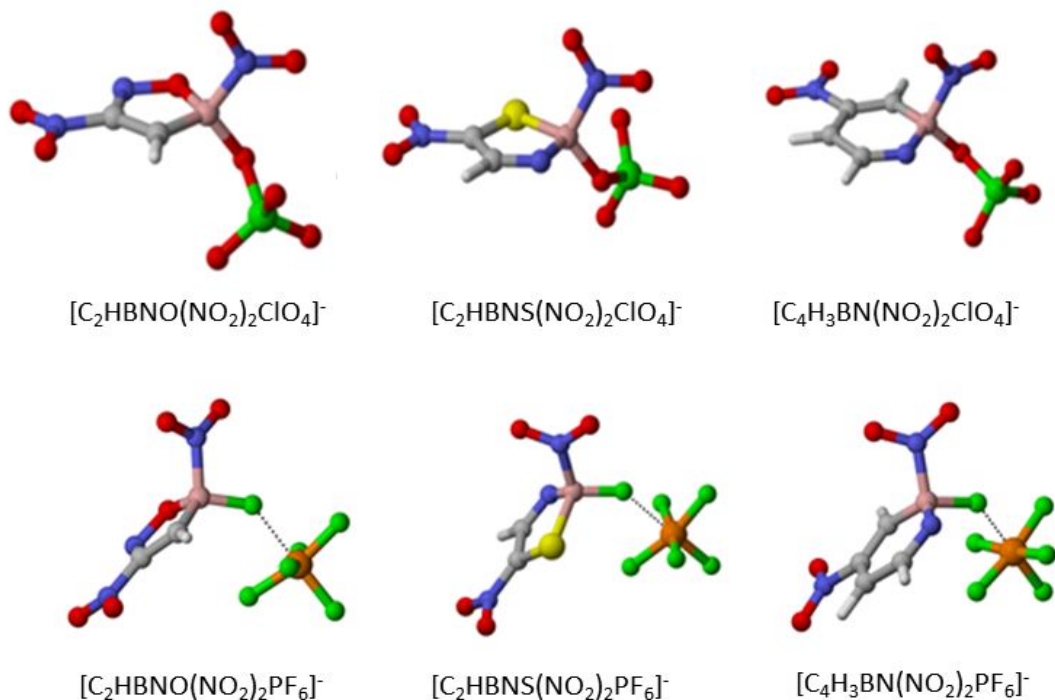


Figure 4. Optimized geometries of ClO₄⁻ and PF₆⁻ trapped by NO₂ substituted Superhalogens

The allied binding energy (BE) values of these superhalogen complexes are tabulated in Tables 2 and 3 respectively.

Table 2. Binding energies of ClO₄⁻ and PF₆⁻ anions trapped by CN substituted Superhalogens (SH) in gas phase and solvent phase.

S. No	Molecules	Binding Energy (Kcal/mol)	
		$(E_{Complex}^-) - (E_{ClO_4^-/PF_6^-} + E_{SH}^0)$	
		(In Gas Phase)	(In Solvent)

			Phase)
1	[TPFPB(CIO ₄)] ⁻	-42.61	-18.49
2	[C ₂ HBNO(CN) ₂ CIO ₄] ⁻	-41.11	-21.25
3	[C ₂ HBNS(CN) ₂ CIO ₄] ⁻	-41.09	-21.92
4	[C ₄ H ₃ BN(CN) ₂ CIO ₄] ⁻	-34.93	-15.17
5	[TPFPB(PF ₆)] ⁻	-25.32	-3.18
6	[C ₂ HBNO(CN) ₂ PF ₆] ⁻	67.82	-6.53
7	[C ₂ HBNS(CN) ₂ PF ₆] ⁻	68.72	-6.87
8	[C ₄ H ₃ BN(CN) ₂ PF ₆] ⁻	76.44	-2.93

Table 3. Binding energies of ClO₄⁻ and PF₆⁻ anions trapped by NO₂ substituted Superhalogens (SH) in gas phase and solvent phase.

S. No	Molecules	Binding Energy (Kcal/mol) $(E_{\text{Complex}}^-) - (E_{\text{ClO}_4^-/\text{PF}_6^-} + E_{\text{SH}}^0)$	
		(In Gas Phase)	(In Solvent Phase)
1	[TPFPB(CIO ₄)] ⁻	-42.61	-18.49
2	[C ₂ HBNO(NO ₂) ₂ CIO ₄] ⁻	-51.17	-31.20
3	[C ₂ HBNS(NO ₂) ₂ CIO ₄] ⁻	-53.86	-34.02
4	[C ₄ H ₃ BN(NO ₂) ₂ CIO ₄] ⁻	-40.86	-22.04
5	[TPFPB(PF ₆)] ⁻	-25.32	-3.18
6	[C ₂ HBNO(NO ₂) ₂ PF ₆] ⁻	-34.56	-16.06
7	[C ₂ HBNS(NO ₂) ₂ PF ₆] ⁻	-37.55	-19.26
8	[C ₄ H ₃ BN(NO ₂) ₂ PF ₆] ⁻	-26.25	-8.34

The calculated EA value of the boron-based superhalogens as shown in Tables 1 is high which transpires that the B-center in those molecules is electron deficient and the systems,

therefore, exhibit Lewis acidity as well. Such an insight gains conspicuous the ground upon the estimation of the binding energy values of the complexes with the superhalogens and the ClO_4^- and PF_6^- ions. A negative BE value for all the systems foretells a favorable Lewis acid-base complex formation between the superhalogens and the ClO_4^- and PF_6^- ions. So, these individual superhalogens can be used as anion receptor in Li-ion battery. Although there exist positive binding energies of anion trapped geometries for CN substituted superhalogen molecule in gas phase, note the negative binding energies in solvent phase which render that these molecules form a stable complex with the anions in solvent phase. Apart from this we have further calculated the change in free energy due to binding between the anions and the SHs. This calculation certainly considers entropic contribution in the binding process. The negative free energy changes clearly infer the feasibility of formation of the complexes. The computed ΔG values are given in the Supporting Information (S10). Additionally, it may be noted that we have not used any explicit solvent in our study however, the solvent effect has been introduced by employing polarizable continuum method, which is a common well-known practice to observe the solvent effect on such type of systems while treating quantum mechanically. Since, dispersion is likely to be an important component of binding, wB97XD functional, which uses Grimme's D2 dispersion model has been used in this study.²³ Moreover, we have used dispersion corrected functional like M06-2X²⁸ for few calculations and

found that the qualitative trend observed from the data are similar for both the cases. The results obtained from M06-2X functional have displayed in S11 in Supporting Information. The qualitative trend observed from the data are similar for both the cases. Now, in an attempt to mimic the structure of TFPBP we have modeled analogous tri-substituted boron derivatives using the superhalogens $C_2HBNO(CN)_2$, $C_2HBNS(CN)_2$ and $C_4H_3BN(CN)_2$ in place of the C_6F_5 ligand. The optimized geometries of the $B(SH)_3$ (SH = superhalogen) systems and the NPA charges on the allied B-centers in Figure 5 clearly depict an electron-deficient nature on the boron site and hence an overall Lewis acidity of the molecules.

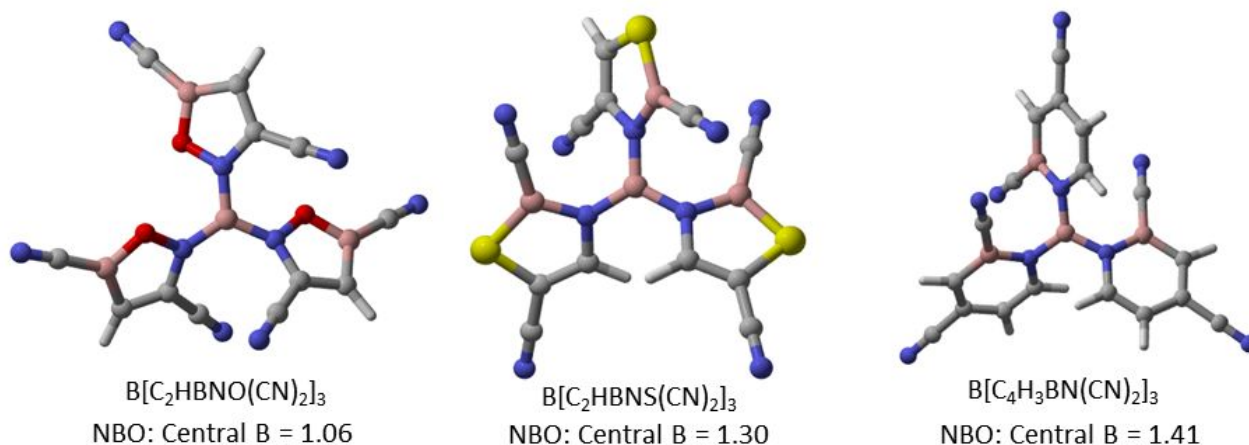


Figure 5. Optimized geometries of trisubstituted B Lewis acid complex with NPA charge of central boron atom.

The binding energies of these anions with TFPBP are also shown in the Table 4 for comparison. It is expected that the newly designed $B(SH)_3$ molecules will act as better anion

receptors than TFPB owing to the presence of a higher formal positive charge on the B-centers. To validate our assumption, we have modeled similar Lewis acid-base complexes of $B(SH)_3$ and the ClO_4^- / PF_6^- anions.

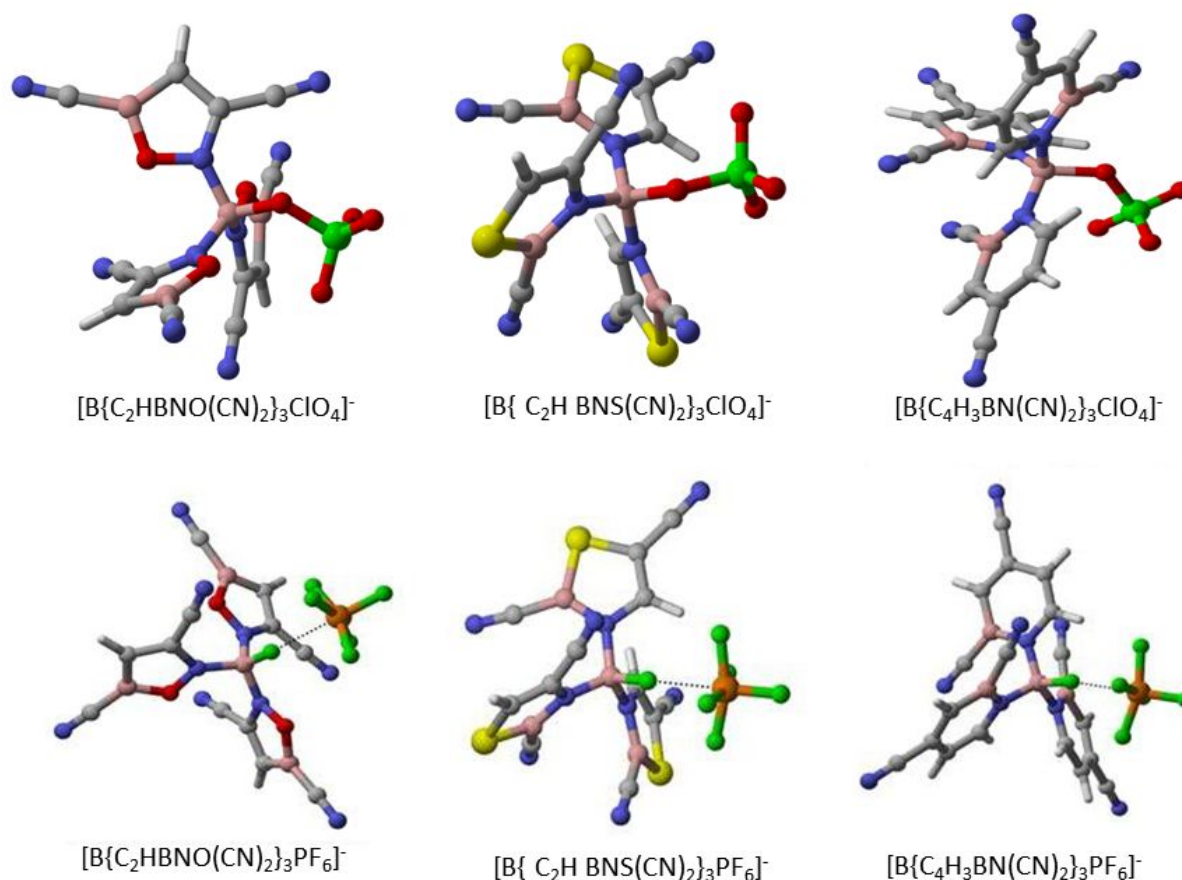


Figure 6. Optimized geometries ClO_4^- and PF_6^- anion trapped by tri-substituted B complex with their binding energies.

Table 4. The comparison of binding energy of newly designed B-tri substituted molecule and TFPB with anions like ClO_4^- , PF_6^-

S. No	Molecules	Binding Energy (Kcal/mol)	
		$(E_{Complex^-}) - (E_{ClO_4^-/PF_6^-} + E_{B(SH_3)^0})$	
		(In Gas Phase)	(In Solvent Phase)
1	[TPFPB(ClO ₄)] ⁻	-42.61	-18.49
2	[B{C ₂ H ₂ NO(CN) ₂ } ₃ ClO ₄] ⁻	-59.24	-28.01
3	[B{C ₂ H ₂ BNS(CN) ₂ } ₃ ClO ₄] ⁻	-55.37	-24.16
4	[B{C ₄ H ₃ BN(CN) ₂ } ₃ ClO ₄] ⁻	-57.03	-21.92
5	[TPFPB(PF ₆)] ⁻	-25.32	-3.18
6	[B{C ₂ H ₂ NO(CN) ₂ } ₃ PF ₆] ⁻	-42.64	-13.47
7	[B{C ₂ H ₂ BNS(CN) ₂ } ₃ PF ₆] ⁻	-42.42	-12.93
8	[B{C ₄ H ₃ BN(CN) ₂ } ₃ PF ₆] ⁻	-43.87	-10.63

The ground state geometries and the binding energy values of the trisubstituted B-complexes in gas as well as in solvent phases are shown in Figure 6 and Table 4 respectively. Clearly, the low binding energy values establish the fact that the newly designed superhalogen-coordinated Lewis acids can function as better anionic receptors than TPFPB.

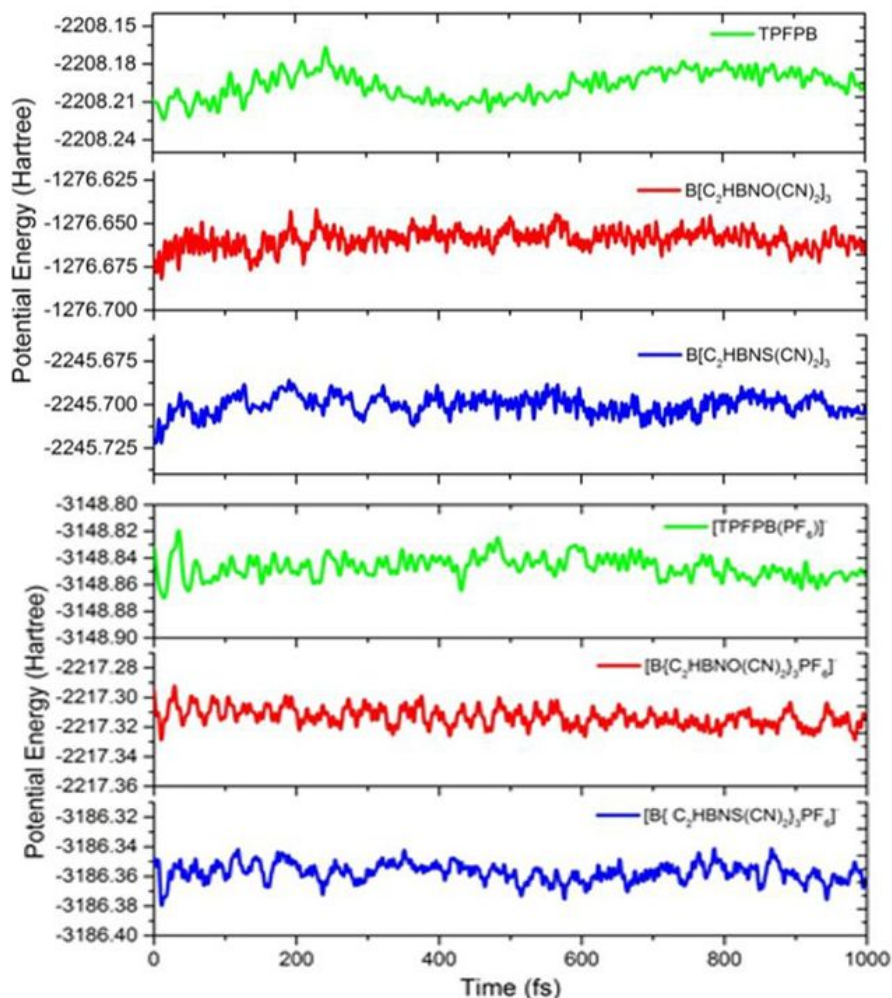


Figure 7. ADMP Molecular Dynamic simulation at room temperature of all studied molecules comparison with TPFPB.

The respective calculations for the NO_2 substituted superhalogen have just been omitted to avoid the plausible safety issue arising out of the presence of too many NO_2 groups in the system. To observe the dynamical stability of the above-studied molecules we have further performed an Atom Centered Density Matrix Propagation (ADMP) molecular dynamics simulations to generate trajectory. We have generated a 1000 fs long trajectory at room

temperature with 1 fs time interval. As mentioned earlier, the gas phase optimized geometries were considered as initial starting structures for the simulations. The time evolution of potential energies of the molecules are shown in Figure 7. The figure infers that there is no such deviation in energy profile. This further suggests that the molecules are stable at room temperature. The geometries of the compounds as obtained from the simulated trajectories at different time interval have been shown in S12 of Supporting Information.

VALIDATION OF THE FORMATION OF DESIGNED ADDITIVES

So far, our calculations have suggested that all the modeled compounds and their TFPB analogues can act as potential additives in Li-ion battery electrolytes. Validation of the formation of modeled compounds included reproduction of available experimentally reported absolute magnetic shielding values and chemical shifts values for different boron-complexes.

Table 5. Absolute magnetic shielding values (ppm) and Chemical Shift (ppm) for different boron complexes calculated at wB97XD/6-311+G(d,p) and B3LYP/6-311+G(2d,p) level of theories.

Molecules	Atom (Centre)	Absolute shielding ($\sigma^{11}\text{B}$, ppm)		Chemical Shift ($\delta^{11}\text{B}$, ppm)	
		Wb97xd/ 6-311+G(d,p)	B3LYP/ 6-311+G(2d,p)	Wb97xd/ 6-311+G(d,p)	B3LYP/ 6-311+G(2d,p)

BF ₃ -OEt ₂	B	99.44	99.44		
TPFPB	B	49.64	45.18	49.8	54.26 (57.0 – 60.0)*
H ₃ P-TPFPB	B	125.84	118.38	-26.4	-18.94 (-17)*
C ₅ H ₃ (CH ₃) ₂ N-TPFPB	B	111.86	107.32	-12.42	-7.88 (-4.7)*
[TPFPB(ClO ₄)] ⁻	B	109.77	104.25	-10.33	-4.81
[TPFPB(PF ₆)] ⁻	B	101.66	96.16	-2.22	3.28
B[C ₂ HBNO(CN) ₂] ₃	B	87.78	84.23	11.66	15.21
[B{C ₂ HBNO(CN) ₂ } ₃ ClO ₄] ⁻	B	109.78	104.05	-10.34	-4.61
[B{C ₂ HBNO(CN) ₂ } ₃ PF ₆] ⁻	B	108.98	103.11	-9.54	-3.67
B[C ₂ HBNS(CN) ₂] ₃	B	75.81	71.17	23.63	28.27
[B{C ₂ HBNS(CN) ₂ } ₃ ClO ₄] ⁻	B	105.33	100.14	-5.89	-0.7
[B{C ₂ HBNS(CN) ₂ } ₃ PF ₆] ⁻	B	104.60	98.71	-5.16	0.73
B[C ₄ H ₃ BN(CN) ₂] ₃	B	65.82	60.56	33.62	38.88
[B{C ₄ H ₃ BN(CN) ₂ } ₃ ClO ₄] ⁻	B	101.21	95.19	-1.77	4.25
[B{C ₄ H ₃ BN(CN) ₂ } ₃ PF ₆] ⁻	B	99.92	94.32	-0.48	5.12

*Experimental chemical shift of the complexes.

$\sigma(^{11}\text{B})$ data are converted to $\delta^{11}\text{B}$ data by $\delta^{11}\text{B} = \sigma(^{11}\text{B}) (\text{BF}_3\text{-OEt}_2) - \sigma(^{11}\text{B}) + 0$, with

$\sigma(^{11}\text{B}) (\text{BF}_3\text{-OEt}_2) = 99.44$ and $\delta^{11}\text{B} (\text{BF}_3\text{-OEt}_2) = 0$.

In this context, it may be noted that till now our proposed compounds lack direct experimental evidences. Therefore, validation of the theoretically calculated data for the designed compounds involves an indirect approach. In this approach, our theoretically calculated data for known boron compounds like TPFPB has been verified with the available experimental ^{11}B NMR

1 data. As reported experimentally the ^{11}B spectra of TFPB to external $\text{BF}_3\cdot\text{Et}_2\text{O}$ appears near
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4 57-60^{29, 30} ppm region. This is in good agreement with the theoretically predicted ^{11}B chemical
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7 shift value of the compound TFPB as shown in Table 5. Further, it is evident from the
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10 experimental works that in general ^{11}B chemical shifts show large changes on complexation, e.
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13 g, the shifts become -17 and -4.7 ppm for the complex H_3P -TFPB and $\text{C}_5\text{H}_3(\text{CH}_3)_2\text{N}$ -TFPB^{29,30},
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16 respectively. For validation purpose, we have additionally calculated the ^{11}B chemical shift
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19 value for these two complexes, and find that the result matches excellently with the
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22 experimental values (see Table 5). It can be further noted from Table 5 that the calculated ^{11}B
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25 chemical shift values show large changes for all the proposed complexes. This is in agreement
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28 with the general trend of ^{11}B chemical shift values of the proposed boron complexes. Our all
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31 such findings infer that the proposed models are potentially strong to reproduce experimental
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44 CONCLUSIONS

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47 In conclusion, we have successfully designed B based Lewis acid with organic heterocyclic
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50 aromatic superhalogens. Not only the geometry but also the properties and anion binding
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53 ability of designed complexes are at par with TFPB. In fact, in terms of anion binding energy,
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56 the designed Lewis acids act better than TFPB. However, so far we have not studied the
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influences of these additives on the conductivity and the related electrochemical studies. Considering its importance, we are presently working on such aspects in our laboratory. However, we believe that our findings, as described in this work will definitely open door to the experimentalist to synthesize a new Lewis acid which can act as better additives in Li-ion battery electrolytes.

SUPPORTING INFORMATION

Optimized Cartesian coordinates of all studied systems (Figure S1-S9), computed free energy change (S10), results obtained by using M06-2X functional (S11), along with snapshot of anion trapping complex at room temperature in different time interval (S12) are given in supporting information.

ACKNOWLEDGMENT

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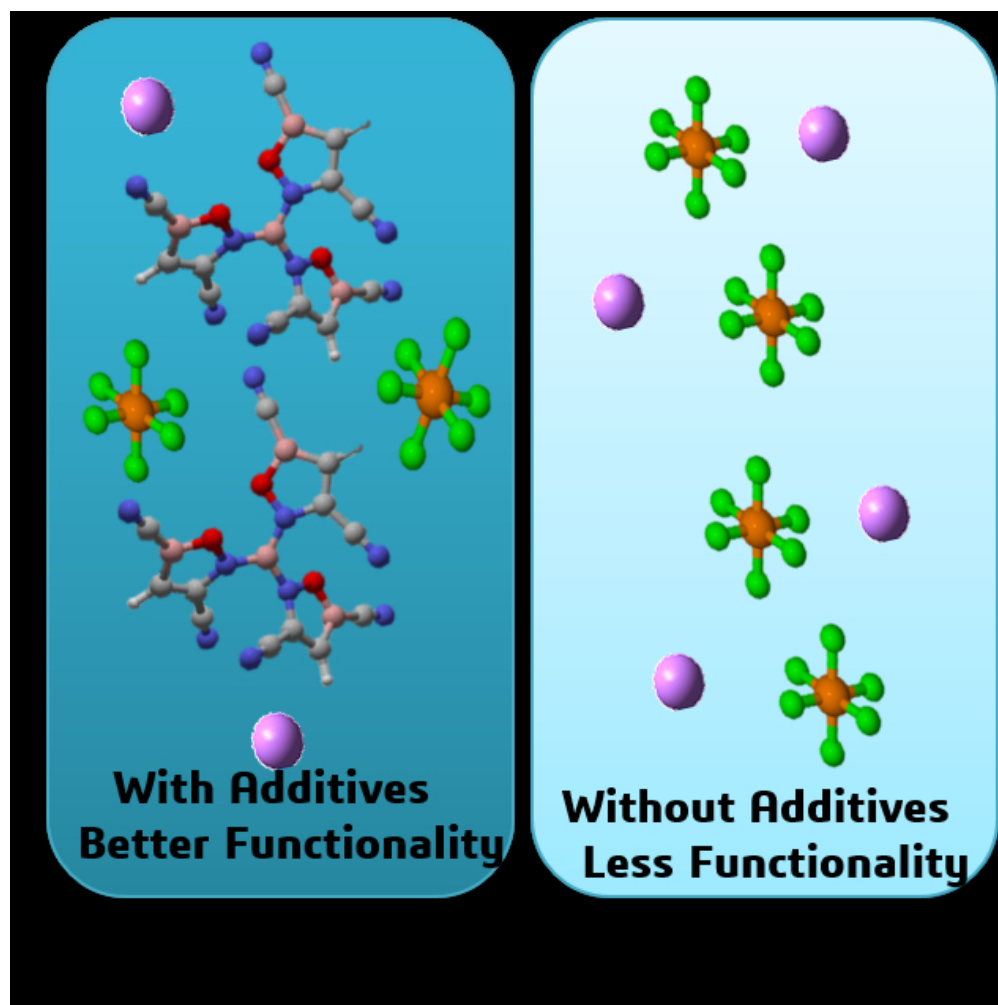
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TOC Graphic

107x107mm (150 x 150 DPI)